

KRISHNA Koushik Thokala

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Summary

Cloud Infrastructure Engineer with 2 years of 24/7 production operations experience supporting Tier-1 carrier networks and hands-on AWS/EKS platform engineering. Built Terraform-managed Kubernetes and serverless systems with GitOps delivery, observability, autoscaling, rollback validation, least-privilege access, network boundaries, and cost-controlled AWS teardown.

Technical Skills

Cloud & AWS: AWS, VPC, IAM, EC2, EKS, ECR, ALB, Lambda, API Gateway, DynamoDB, RDS, S3, CloudWatch, Secrets Manager, KMS
Infrastructure & Platform: Terraform, Docker, Kubernetes, Helm, Argo CD, GitOps, GitHub Actions, CI/CD, Linux, Bash
Security & Access: IAM Policies, Trust Policies, OIDC, IRSA, RBAC, NetworkPolicies, Security Groups, API Keys, CORS
Observability & Reliability: Prometheus, Grafana, CloudWatch Logs/Metrics/Alarms, HPA, k6 Load Testing, SLO/SLA, MTTR
Networking & Programming: TCP/IP, DNS, BGP/MPLS, Load Balancing, Python, PostgreSQL, FastAPI, React
Certifications: AWS Certified Solutions Architect – Associate (2026)

Experience

Tata Communications

July 2022 – July 2024

Senior Engineer, Shift Lead (24/7 production NOC operations)

Pune, India

- Led 24/7 production NOC shift operations for 25-30 Tier-1 carrier clients, coordinating incident triage, escalation, field dispatch, TAC/vendor follow-up, and SLA-driven service restoration.
- Diagnosed high-severity BGP/MPLS, DWDM, and transport-network failures across Juniper, Huawei, Ciena, and Alcatel platforms using alarms, logs, loop testing, and segment-wise fault isolation.
- Led 5-person on-call shifts handling 40+ daily priority incidents/escalations, including structured handoffs, customer-premises testing, RCA notes, and cross-functional follow-up through resolution.
- Supported 99.9% annual network availability and 4-hour SLA restoration targets through incident command, vendor coordination, field-team execution, and RCA follow-up.

Projects

CloudOps GitOps Platform | EKS, Argo CD, Helm, Terraform, GitHub Actions

GitHub

- Designed an EKS GitOps platform with 4 Argo CD Applications for dev/staging/prod/observability, Git-managed Helm values, namespace boundaries, ResourceQuotas, RBAC, NetworkPolicies, and restricted pod security settings.
- Provisioned 20 AWS resources with Terraform and demonstrated PR-style image promotion, drift self-healing, Git revert rollback, Prometheus/Grafana observability, AWS Budgets, and Terraform destroy validation.

CloudOps SRE Platform | EKS, Terraform, Helm, FastAPI, React, RDS

GitHub

- Built an EKS-based SRE platform for service health, deployment history, incident timelines, MTTR, SLO tracking, and error budget views with FastAPI, React, PostgreSQL, and Alembic migrations.
- Deployed workloads behind ALB Ingress with Helm-managed probes, resource limits, Secrets Manager-backed credentials, request IDs, structured logs, CloudWatch visibility, and RDS access restricted by security groups.
- Measured HPA scale-out under k6 load as backend replicas scaled from 2 to 6 pods (444% CPU vs 60% target), processing 2,035 requests at 99.5% success with 1.52s p95 latency.

CloudOps Uptime Monitor | Lambda, DynamoDB, API Gateway, Terraform

GitHub

- Built a serverless uptime monitor with EventBridge-scheduled Lambda checks, DynamoDB history/latest-status tables, SNS state-change alerts, 9 CloudWatch metrics, and a CloudFront-hosted React dashboard.
- Hardened dashboard APIs with API keys, throttling, restricted CORS, KMS, DLQs, SSRF protection, unsafe URL blocking, and CI gates; optimized current-status reads from 86,400 records to 10 latest-status rows.
- Executed an outage/recovery drill with 1-second failure detection, 26-second dashboard visibility, recovery notification, and zero duplicate alerts during sustained downtime.

Education

Indiana University Bloomington

August 2024 – May 2026

Master of Science in Computer Science, GPA: 3.5/4.0

Bloomington, IN

SRM Institute of Science and Technology

June 2018 – June 2022

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.43/10

Chennai, India